Routing algorithms: shortest path routing, Flooding, Hierarchical routing, Broadcast, Multicast, distance vector routing, Link State Routing

ROUTING ALGORITHMS

The routing algorithm is that part of the network layer software responsible for deciding which output line an incoming packet should be transmitted on.
Router as having two processes inside it. One of them handles each packet as it arrives, looking up the outgoing line to use for it in the routing tables. This process is forwarding. The other process is responsible for filling in and updating the routing tables. That is where the routing algorithm comes into play.

Routing algorithms can be grouped into two major classes:

1. Nonadaptive algorithms
2. Adaptive algorithms

Nonadaptive algorithms do not base their routing decisions on any measurements or estimates of the current topology and traffic. Instead, the choice of the route to use to get from I to J (for all I and J) is computed in advance, offline, and downloaded to the routers when the network is booted. This procedure is sometimes called static routing algorithms.

Adaptive algorithms, in contrast, change their routing decisions to reflect changes in the topology, and sometimes changes in the traffic as well. These dynamic routing algorithms.

The Optimality Principle

It states that if router J is on the optimal path from router I to router K, then the optimal path from J to K also falls along the same route. To see this, call the part of the route from I to Jr 1 and the rest of the route r 2. If a route better than r 2 existed from J to K, it could be concatenated with r 1 to improve the route from I to K, contradicting our statement that r 1r 2 is optimal.
set of optimal routes from all sources to a given destination form a tree rooted at the destination. Such a tree is called a sink tree

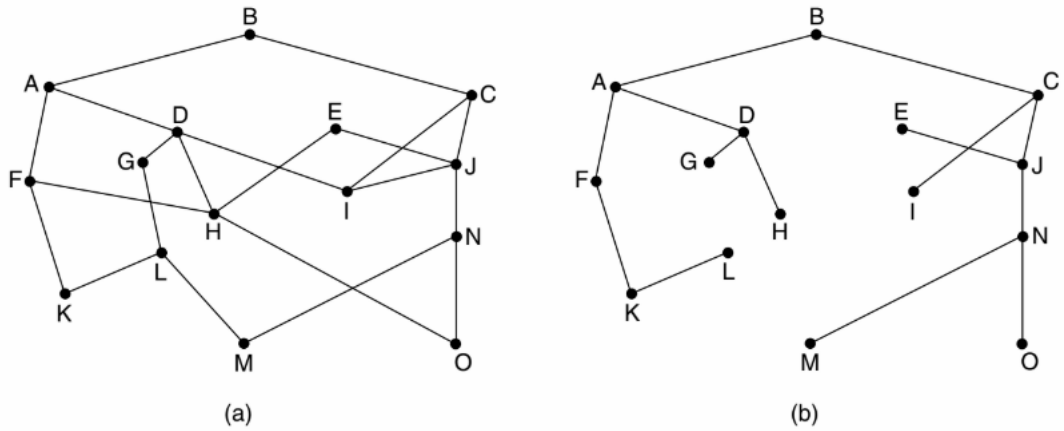


Figure 5-6. (a) A network. (b) A sink tree for router *B*.

Shortest Path Algorithm:

The Shortest Path Algorithm computes optimal paths when the complete network topology and link costs are known. These optimal paths are the paths that a distributed routing algorithm aims to find.

The idea is to build a graph of the network, with each node of the graph representing a router and each edge of the graph representing a communication line, or link. To choose a route between a given pair of routers, the algorithm just finds the shortest path between them on the graph.

One way of measuring **path length** is by counting the **number of hops**, where each hop represents one link between two routers. Another metric is the **geographic distance in kilometers**, where the path with the shortest physical distance may be preferred. A third metric is **network delay**, where each edge is labeled with the **mean delay of a standard test packet**, measured through periodic or hourly test runs. Thus, the optimal path can be determined based on the metric being used, such as hops, distance, or delay.

Several algorithms for computing the shortest path between two nodes of a graph are known. This one is due to Dijkstra (1959) and finds the shortest paths between a source and all destinations in the network.

Dijkstra's Shortest Path Algorithm

Dijkstra's algorithm, proposed by Edsger Dijkstra in 1959, is used to find the shortest paths from a source node to all other nodes in a network. Each node is assigned a label representing the distance from the source along the best path known so far. Initially, no paths are known, so the source is assigned a distance of **0** and all other nodes are assigned infinity **(∞)**. As the algorithm explores the network, these labels are updated whenever a shorter path is found. A label can be tentative or permanent.

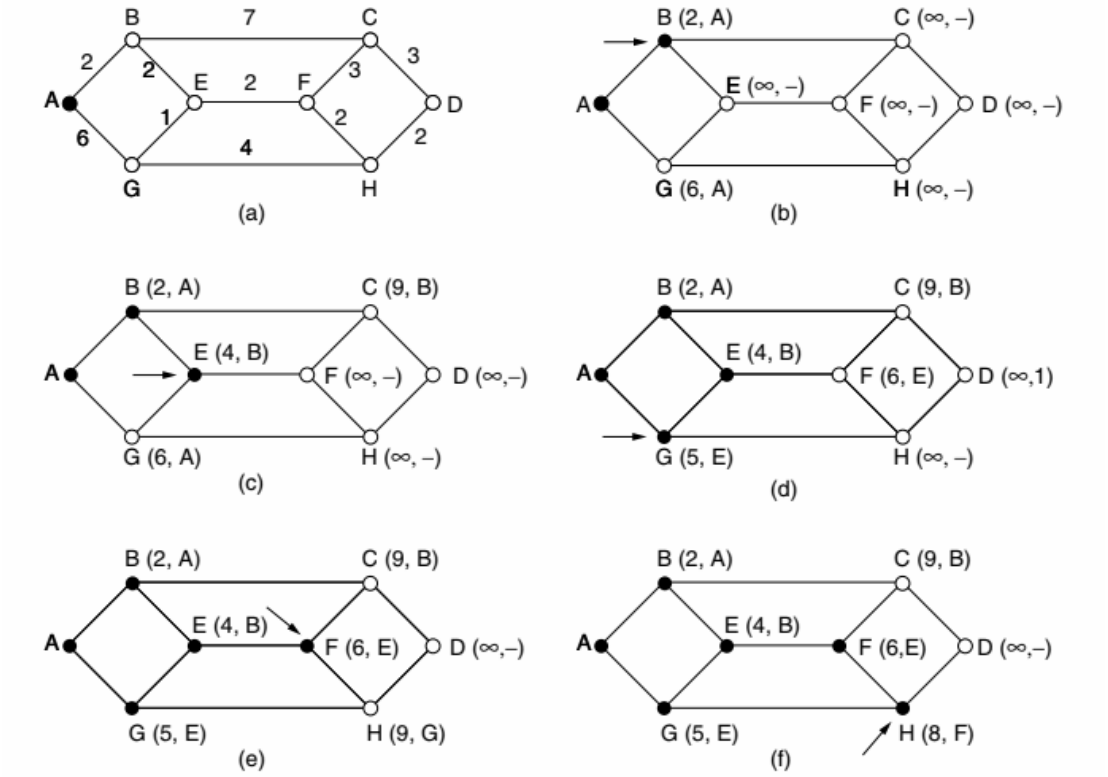


Figure 5-7. The first six steps used in computing the shortest path from *A* to *D*. The arrows indicate the working node.

To choose a route between a given pair of routers, the algorithm just finds the shortest path between them on the graph

Step 1. Start with the local node (router) as the root of the tree. Assign a cost of 0 to this node and make it the first permanent node.
Step 2. Examine each neighbor of the node that was the last permanent node.
Step 3. Assign a cumulative cost to each node and make it tentative
Step 4. Among the list of tentative nodes

a. Find the node with the smallest cost and make it Permanent
b. If a node can be reached from more than one route then select the route with the shortest cumulative cost.

Step 5. Repeat steps 2 to 4 until every node becomes permanent

Example: To find shorted path from A to D in the above diagram

1. **Start from source node A** and mark A as **permanent**. Examine all nodes **directly connected to A**. Assign each adjacent node a **tentative distance** equal to the distance from A. Along with the distance, record the **previous/probe node** from which the node was reached. This helps reconstruct the final shortest path.
2. After examining A's neighbors, look at **all tentatively labeled nodes** in the graph. Select the node with the **smallest tentative distance** and make it **permanent**. This becomes the new **working node**.
3. Examine all nodes adjacent to the new working node. For each adjacent node, calculate:
   New distance = distance of working node + distance of the connecting edge
   a. If the new distance is **smaller than the node's current tentative distance**, update its label and record the working node as its predecessor.

   b. Again, search the entire graph for the **tentatively labeled node with the smallest value**. Make that node **permanent** and use it as the new working node.
4. Repeat this process until **D becomes permanent** (or until all nodes have been made permanent if shortest paths to every destination are required).
5. Finally, use the recorded **predecessor/probe nodes** to trace backward from D to A and obtain the **shortest path from A to D**.

Flooding in Routing

Flooding is a simple routing technique in which each router makes decisions using only its **local knowledge** of the network. When a router receives a packet, it **forwards a copy of the packet through every outgoing link except the link on which the packet arrived**. This ensures that the packet reaches many or potentially all parts of the network, but it also creates many **duplicate packets**.

The main problem with flooding is that the number of duplicate packets can become extremely large and may even become **infinite** if there is no mechanism to stop them. Two common methods are used to control flooding:

1. **Hop Counter (TTL):**
   - A hop counter is placed in the packet header.
   - The counter is decreased by 1 every time the packet passes through a router.
   - When it reaches **0**, the packet is discarded.
   - If the exact path length is unknown, the counter can be set to the **network diameter** (the maximum possible distance between two nodes).
2. **Sequence Numbers:**
   - The source router assigns a **sequence number** to each packet.
   - Each router keeps track of the sequence numbers of packets it has already seen from each source.
   - If a router receives a packet with a sequence number that it has already seen, it **discards it instead of flooding it again**.
   - This is more efficient because it prevents the same packet from being forwarded repeatedly.

Hierarchical Routing

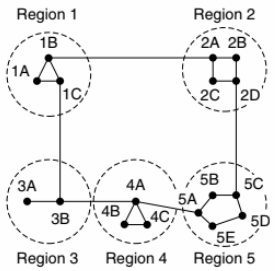
Hierarchical routing is a routing technique in which a large computer network is divided into **smaller groups or regions called areas**. Routing is performed at different levels of the hierarchy. This reduces the size of routing tables and makes routing more efficient.

How it works:

1. **Division into regions:** The network is divided into areas, zones, or regions.
2. **Routing within an area:** Routers maintain detailed information about destinations **inside their own area**.
3. **Routing between areas:** For destinations in another area, routers use summarized information and forward packets to a suitable router in that area.
4. **Multiple levels:** Large networks can have several levels, such as **local area → regional area → backbone**.
5. **Route aggregation:** Several networks can be represented by a single routing entry, reducing routing-table size.

Example: Consider a network divided into **Area A, Area B, and Area C**. A router in Area A does not need to know every individual network in Area B. It only needs to know **which router/area leads to Area B**.

Example 2 : A quantitative example of routing in a two-level hierarchy with five regions. The full routing table for router 1A has 17 entries, as shown in Fig. 5-14(b). When routing is done hierarchically, as in Fig. 5-14(c), there are entries for all the local routers, as before, but all other regions are condensed into a single router, so all traffic for region 2 goes via the 1B-2A line, but the rest of the remote traffic goes via the 1C-3B line. Hierarchical routing has reduced the table from 17 to 7 entries.



(a)

Full table for 1A

Dest.	Line	Hops
1A	–	–
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

(b)

Hierarchical table for 1A

Dest.	Line	Hops
1A	–	–
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

(c)

Advantages: **Reduces routing table size** and **Reduces routing overhead**

Disadvantage: The route may not always be the **shortest possible route**, because routers use summarized information between different hierarchy levels.

In short: Hierarchical routing divides a large network into smaller areas and uses different levels of routing to make routing **scalable, efficient, and manageable**.

Broadcast routing

Broadcast routing is a routing technique in which a packet is sent from **one source node to all other nodes** in a network. It is mainly used when the sender needs to distribute the same information to multiple or all nodes.

Methods of Broadcast Routing

1. Flooding:

Flooding is a simple broadcast routing method in which a router forwards the received packet to **all its neighboring routers**, except the router from which the packet was received. This process continues until the packet reaches all routers in the network. A **hop count or sequence number** can be used to prevent packets from circulating indefinitely. Although flooding is reliable, it can generate many duplicate packets and cause heavy network traffic.

2. Multidestination Routing:
In multidestination routing, a single packet contains a **list of multiple destination addresses**. When a router receives the packet, it examines the destination list and forwards copies through the appropriate outgoing links. Destinations that can be reached through the same link can be grouped together. This method reduces

the need to send a separate packet to every destination, but routers still need to manage multiple destination information.

3. Spanning Tree:
In the spanning-tree method, a **spanning tree connecting all routers** in the network is created. The broadcast packet is forwarded only along the links of this tree. Since a spanning tree does not contain loops, every router receives the broadcast packet **only once**, avoiding duplicate packets. This method is more efficient than flooding and reduces unnecessary network traffic.

4. Reverse Path Forwarding (RPF):

When a broadcast packet arrives at a router, the router checks to see if the packet arrived on the link that is normally used for sending packets toward the source of the broadcast. If so, there is an excellent chance that the broadcast packet itself followed the best route from the router and is therefore the first copy to arrive at the router. This being the case, the router forwards copies of it onto all links except the one it arrived on. If, however, the broadcast packet arrived on a link other than the preferred one for reaching the source, the packet is discarded as a likely duplicate.

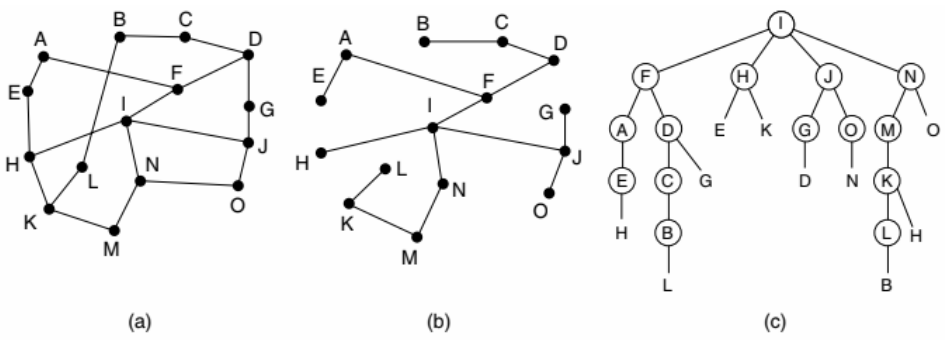


Figure 5-15. Reverse path forwarding. (a) A network. (b) A sink tree. (c) The tree built by reverse path forwarding.

Sink Tree

A **sink tree** is a tree-shaped routing structure formed with a particular **destination node (sink) as the root**. It shows the **shortest path from every other router to that particular destination**. Each router selects the next router that lies on the shortest path toward the sink, so packets can be forwarded toward the destination without loops.

Multicast Routing

Multicast routing is a routing technique in which a packet is sent from **one source to a specific group of destination hosts**, rather than to every host in the network. The hosts that want to receive the multicast packets join a **multicast group**. Routers use multicast routing information to create efficient paths from the source to all members of the group.

1. MOSPF – Multicast OSPF

MOSPF (Multicast Open Shortest Path First) is a multicast routing protocol based on **OSPF (Open Shortest Path First)**. It uses the link-state information maintained by OSPF to build a **shortest-path tree** from the multicast source to the members of a multicast group. When a multicast packet arrives, the router calculates the shortest path toward the group members and forwards copies only through the required interfaces. MOSPF uses **IGMP group membership information** to know which networks contain multicast receivers. It is efficient in large networks but requires additional processing and memory to calculate multicast routes.

2. DVMRP – Distance Vector Multicast Routing Protocol

DVMRP (Distance Vector Multicast Routing Protocol) is one of the earliest multicast routing protocols. It is based on the **distance-vector routing approach** and uses a technique called **Reverse Path Forwarding (RPF)**. When a multicast packet arrives, the router checks whether it came through the correct interface for reaching the source. If it did, the router forwards the packet to other interfaces; otherwise, it discards it. DVMRP initially **floods** multicast packets throughout the network and then uses a **pruning process** to remove branches that do not contain multicast group members.